

day02笔记

一、作业

1、从“城市-区域”字段中拆分两级信息

数据列：`locations = ["北京市-海淀区", "上海市-浦东新区", "广州市-天河区"]`

拆分为两个列表：

→ `cities = ["北京市", "上海市", "广州市"]`

→ `districts = ["海淀区", "浦东新区", "天河区"]`

```
locations = ["北京市-海淀区", "上海市-浦东新区", "广州市-天河区"]
# 市
cities = []
# 区
districts = []
# 遍历操作
for loc in locations:
    # 北京市-海淀区
    if '-' in loc:
        # 拆包操作
        c, d = loc.split('-')
        cities.append(c)
        districts.append(d)

print(cities)
print(districts)

print('-' * 20)

# 列表
# lis = ['张三', '李四', '坤坤']
# print(lis[0], lis[1], lis[2])

# name1, name2, name3 = ['张三', '李四', '坤坤']
# print(name1, name2, name3)

# 元祖
# t1, t2 = ('张三', '李四')
# print(t1, t2)

# 字符串
# n1, n2, n3 = '123'
# print(n1, n2, n3)
```

2、统计字符串的个数

```
text = 'abcdffqdafig'
"""
```

```

counter = {
    a: 2,
    b: 1
}
"""

counter = {}
# 遍历文本数据
for char in text:
    # 判断下某个字符是否在字典中存在，把字符串中的字符当成字典中的键来使用 key: value 形式（键值对）
    # char ---> 'a'
    # char ---> 'b'
    # ...
    if char in counter:
        # 说明已经存在
        # counter[char] ---> 值是数字
        counter[char] += 1
    else:
        # 说明不存在，把这个字符串当成字典的键添加进字典，并赋值为1
        counter[char] = 1

print(counter)

# print(counter['a'])
# print(counter['b'])
# print(counter.get('a'))

```

二、字符串练习题

1、从“姓名(英文)”字段提取英文名

数据: `names = ["张三(Tom)", "李四(Lucy)", "王五(Bob)"]`

提取英文名: `['Tom', 'Lucy', 'Bob']`

1、从“姓名(英文)”字段提取英文名

数据: ``names = ["张三(Tom)", "李四(Lucy)", "王五(Bob)"]``

提取英文名: ``['Tom', 'Lucy', 'Bob']``

"""

`names = ["张三(Tom)", "李四(Lucy)", "王五(Bob)"]`

`english_names = []`

`for name in names:`

`if '(' in name and ')' in name:`

`# find() 查找字符串某个字符的位置，返回的是索引`

`# start 表示的就是提取字符串里面字符的开始位置`

`start = name.find('(') + 1`

`# end 表示的就是提取字符串里面字符的结束位置。因为切片的时候，是不包含结束位置的`

`end = name.find(')')`

`en_name = name[start:end]`

`english_names.append(en_name)`

`print(english_names) # ['Tom', 'Lucy', 'Bob']`

2、清洗含单位的数值字段 ("100kg" → 100)

数据: `weights = ["100kg", "200 kg", "50.5kg", "300g", "1.5 t"]`

统一转换为“千克”数值 (假设: 1t = 1000kg, 1g = 0.001kg) :

结果: `[100.0, 200.0, 50.5, 0.3, 1500.0]`

```
# 脏数据
weights = ["100kg", "200 kg", "50.5kg", "300g", "1.5 t"]
result = []

for w in weights:
    # 解决空格问题
    w = w.replace(' ', '')
    # 单位的转换
    if w.endswith('kg'):
        num = float(w[:-2])
        result.append(num)
    elif w.endswith('g'):
        num = float(w[:-1]) * 0.001
        result.append(num)
    elif w.endswith('t'):
        num = float(w[:-1]) * 1000
        result.append(num)

print(result)
```

3、生成数据脱敏后的展示字段 (姓名、手机、邮箱)

原始数据:

```
data = [
    {"name": "张三丰", "phone": "13812345678", "email": "zhang@example.com"},
    {"name": "李莫愁", "phone": "13987654321", "email": "li@example.org"}
]
```

脱敏规则:

- 姓名: 保留姓, 名用 * 代替 → "张**"
- 手机: 138*****5678
- 邮箱: z***@example.com

```
data = [
    {"name": "张三丰", "phone": "13812345678", "email": "zhang@example.com"},
    {"name": "李莫愁", "phone": "13987654321", "email": "li@example.org"}
]

masked_data = []

# d 表示的是列表中的每一个字典
for d in data:
    # 处理姓名
    name = d['name']
```

```

masked_name = name[0] + '*' * (len(name) - 1) if len(name) > 1 else name

# 处理电话号码
phone = d['phone']
masked_phone = phone[:3] + '****' + phone[7:] if len(phone) == 11 else phone

# 处理邮箱地址
email = d['email']
# 邮箱格式的简单验证
if '@' in email:
    local, domain = email.split('@')
    masked_local = local[0] + '***' if len(local) > 1 else local
    masked_email = masked_local + '@' + domain
else:
    masked_email = email

# 添加进列表中
masked_data.append({
    'name': masked_name,
    'phone': masked_phone,
    'email': masked_email
})

print(masked_data)

```

三、三目表达式

- 1、名称：三元运算符、条件表达式
- 2、是对双分支选择结构的简化操作
- 3、语法

```

# x = 10
# y = 20
# if x > y:
#     print(x)
# else:
#     print(y)

# x = 30
# y = 20
# if 前面放的是满足条件的结果，else 后面放的是不满足条件的结果
# result = x if x > y else y
# print(result)

# 根据年龄判断是否成年
age = 20
result = '成年人' if age >= 18 else '未成年人'
print(result)

```

四、数据处理练习题

1、在处理文件路径时，可以使用 `.rfind()` 来查找最后一个斜杠的位置，从而提取文件名。

```
# file_path = "/home/user/documents/report.txt"
# 0 从左边往右边进行查找
# print(file_path.find('/'))
# 20 从右边往左边查找
# print(file_path.rfind('/'))
# 注意点: find() 和 rfind() 如果没有查找找对应的字符的位置会返回-1
# print(file_path.find('-'))
# print(file_path.rfind('-'))

file_path = "/home/user/documents/report.txt"
last_index = file_path.rfind("/")
if last_index != -1:
    file_name = file_path[last_index+1:]
    print(file_name)
else:
    print('没有找到')
```

2、从URL中提取参数 `url = "https://api.example.com/v1/users?name=Alice&age=30&city=Beijing"`

需要提取的数据: `name=Alice&age=30&city=Beijing`

输出结果: `{'name': 'Alice', 'age': '30', 'city': 'Beijing'}`

```
url = "https://api.example.com/v1/users?name=Alice&age=30&city=Beijing"
if '?' in url:
    query_str = url.split('?')[1] # name=Alice&age=30&city=Beijing
    # 把字符串转成字典形式
    params = {}
    # 先进行切割处理
    items = query_str.split('&') # ['name=Alice', 'age=30', 'city=Beijing']
    # 遍历操作
    for it in items:
        # name=Alice
        # age=30
        # city=Beijing
        key, value = it.split('=')

        params[key] = value

    print(params) # {'name': 'Alice', 'age': '30', 'city': 'Beijing'}
```

五、字典和列表组合操作

1、数据

```
school_data = {
    "学校名称": "英才实验中学",
    "年级": "高一年级",
```

```
"班级列表": [
  {
    "班级ID": "G1-01",
    "班级名称": "高一(1)班",
    "班主任": {
      "教师ID": "T001",
      "姓名": "张伟",
      "性别": "男",
      "联系电话": "13800138001",
      "入职年份": 2015,
      "负责科目": ["语文", "德育"]
    },
    "任课教师": [
      {
        "教师ID": "T002",
        "姓名": "李芳",
        "科目": "数学",
        "职称": "高级教师"
      },
      {
        "教师ID": "T003",
        "姓名": "王强",
        "科目": "英语",
        "职称": "一级教师"
      },
      {
        "教师ID": "T004",
        "姓名": "赵敏",
        "科目": "物理",
        "职称": "二级教师"
      }
    ],
    "班级信息": {
      "成立日期": "2024-09-01",
      "最大容量": 45,
      "当前人数": 42,
      "教室编号": "A201",
      "状态": "正常"
    },
    "学生名单": [
      {
        "学号": "S2024001",
        "姓名": "陈小明",
        "性别": "男",
        "出生日期": "2008-03-15",
        "家庭信息": {
          "父亲姓名": "陈建国",
          "母亲姓名": "李秀英",
          "联系电话": "13900139001",
          "家庭住址": "北京市朝阳区XX路XX号"
        },
        "选课情况": ["数学", "物理", "英语", "语文", "体育", "美术"],
        "学期成绩": {
```

```
        "2024秋季学期": {
            "语文": 88,
            "数学": 92,
            "英语": 85,
            "物理": 89,
            "体育": 95,
            "美术": 90
        },
        "2025春季学期": {
            "语文": 90,
            "数学": 94,
            "英语": 88,
            "物理": 91,
            "体育": 96,
            "美术": 92
        }
    },
    "考勤记录": {
        "2024秋季学期": {
            "出勤天数": 92,
            "缺勤天数": 3,
            "迟到次数": 2
        },
        "2025春季学期": {
            "出勤天数": 88,
            "缺勤天数": 1,
            "迟到次数": 0
        }
    },
    "奖惩记录": [
        {
            "时间": "2024-11", "类型": "奖励", "内容": "期中考试年级前十", "颁发人":
"张伟"},
        {
            "时间": "2025-03", "类型": "警告", "内容": "未交作业累计3次", "处理人":
"李芳"}
    ]
},
{
    "学号": "S2024002",
    "姓名": "王丽丽",
    "性别": "女",
    "出生日期": "2008-07-22",
    "家庭信息": {
        "父亲姓名": "王建国",
        "母亲姓名": "赵美玲",
        "联系电话": "13900139002",
        "家庭住址": "北京市海淀区XX街XX号"
    },
    "选课情况": ["数学", "物理", "英语", "语文", "音乐", "信息技术"],
    "学期成绩": {
        "2024秋季学期": {
            "语文": 92,
            "数学": 88,
            "英语": 94,
```

```
        "物理": 85,
        "音乐": 98,
        "信息技术": 96
    },
    "2025春季学期": {
        "语文": 93,
        "数学": 90,
        "英语": 96,
        "物理": 87,
        "音乐": 99,
        "信息技术": 97
    }
},
"考勤记录": {
    "2024秋季学期": {
        "出勤天数": 95,
        "缺勤天数": 0,
        "迟到次数": 1
    },
    "2025春季学期": {
        "出勤天数": 89,
        "缺勤天数": 0,
        "迟到次数": 0
    }
},
"奖惩记录": [
    { "时间": "2024-12", "类型": "奖励", "内容": "校园歌手大赛冠军", "颁发人":
"团委"},
    { "时间": "2025-01", "类型": "奖励", "内容": "编程竞赛一等奖", "颁发人":
"王强"}
]
}

],
{
    "班级ID": "G1-02",
    "班级名称": "高一(2)班",
    "班主任": {
        "教师ID": "T005",
        "姓名": "刘洋",
        "性别": "女",
        "联系电话": "13800138005",
        "入职年份": 2018,
        "负责科目": ["英语", "心理"]
    },
    "任课教师": [
        { "教师ID": "T002", "姓名": "李芳", "科目": "数学", "职称": "高级教师"},
        { "教师ID": "T001", "姓名": "张伟", "科目": "语文", "职称": "高级教师"},
        { "教师ID": "T006", "姓名": "孙浩", "科目": "化学", "职称": "一级教师"}
    ],
    "班级信息": {
```



```

        "成立日期": "2024-09-01",
        "最大容量": 45,
        "当前人数": 44,
        "教室编号": "A202",
        "状态": "正常"
    },
    "学生名单": []
    # 学生数据可按需填充
}
],
"统计信息": {
    "总班级数": 2,
    "总学生数": 42 + 0,  # G1-01有42人, G1-02暂无学生
    "教师总数": 6,
    "最新更新时间": "2025-04-05"
}
}

```

2、题目

```

# 1. 打印高一(1)班班主任姓名
# step1 = school_data['班级列表'][0]['班主任']['姓名']
# print(step1) # 张伟

# 2. 打印陈小明的2025春季学期数学成绩
# step1 = school_data['班级列表'][0]['学生名单'][0]['学期成绩']['2025春季学期']['数学']
# print(step1) # 94

# 3. 打印王丽丽的家庭联系电话
# step1 = school_data['班级列表'][0]['学生名单'][1]['家庭信息']['联系电话']
# print(step1) # 13900139002

# 4. 打印高一(1)班所有任课教师姓名
# step1 = school_data['班级列表'][0]['任课教师']
# for item in step1:
#     # 注意点: 对于字符串来说, 外层如果使用的是双引号, 那么内层必须使用单引号。反之亦然
#     print(f'高一(1)班所有任课教师姓名: {item["姓名"]}')

# 5. 统计陈小明确获得的奖励次数
step1 = school_data['班级列表'][0]['学生名单'][0]['奖惩记录']
# result = []
# for it in step1:
#     if it['类型'] == '奖励':
#         result.append(it)
#
# print(len(result))

# result = [it for it in step1 if it['类型'] == '奖励']
# print(len(result)) # 1

# 6、打印陈小明所有学期各科成绩
# step1 = school_data['班级列表'][0]['学生名单'][0]['学期成绩']
# print(step1)

```

```

step1 = school_data['班级列表'][0]['学生名单'][0]
student = step1['学期成绩']

# 注意点：这个是插件检测的时候，没有检测出来
for key, scores in student.items():
    print(f'{key}')
    for subject, score in scores.items():
        print(f'        {subject}: {score}')

```

3、字典遍历

```

# 字典遍历
# dic = {'name': '张三', 'age': 18, 'gender': '男'}
# 直接遍历字典，拿到的是字典的键
# for key in dic:
#     print(key)

# 遍历字典dic.keys()，拿到的是字典的键
# for key in dic.keys():
#     print(key)

# 遍历字典dic.values()，拿到的是字典的值
# for value in dic.values():
#     print(value)

# 遍历字典dic.items()，拿到的是结果是键和值
# for key, value in dic.items():
#     print(key, value)

```

六、综合练习

编写函数，检测密码强度：至少8位，包含大写字母、小写字母和数字。

输出结果

```

print(check_password("Abc12345")) # True
print(check_password("abc12345")) # 密码缺少：大写字母
print(check_password("ABC12345")) # 密码缺少：小写字母
print(check_password("Abcdefgh")) # 密码缺少：数字
print(check_password("Ab1")) # 密码长度至少为8位
print(check_password("hello")) # 密码长度至少为8位
print(check_password("Helloworld")) # 密码缺少：数字

```

七、作业

1、实现函数统计字符串中元音字母（a, e, i, o, u）的数量

2、商品信息管理

- 新增商品
- 删除商品（根据id删除商品）

- 商品某条信息更改（根据商品id，修改商品名称，价格）

```
product = [  
    {'id': '001', 'name': 'huawei mate80', 'price': 5699, 'count': 2}  
]
```